

Bank Customer Churn Prediction

AIML Mini-Project | Himshikhar Track | Logistic Regression + Random Forest | Comparison Study

1. Problem Statement

Customer churn directly impacts bank revenue — acquiring a new customer costs 5-7x more than retaining an existing one. This project builds a binary classifier to predict whether a bank customer is likely to exit, using demographic and account-usage data. The goal is not just high accuracy but honest evaluation: reporting precision, recall, and explaining specific cases the model misclassifies.

2. Approach & Methodology

Dataset: Bank Customer Churn (Kaggle) — 10,000 customers, 14 features, 20.4% churn rate.

Step	What	How	Why
1. Split	60/20/20 train/val/test	Stratified split	Val set used for threshold tuning; test held out
2. Preprocess	Scale numeric, encode categorical	StandardScaler + OneHotEncoder	No leakage: fit on train only
3. Baseline	Logistic Regression	Default params, max_iter=1000	Floor model to beat
4. Improve	Random Forest + class_weight	200 trees, max_depth=10, balanced w	Handles imbalance without SMOTE
5. Tune	Find optimal threshold	Precision-recall curve on val set	Maximizes F1 vs default 0.5
6. Evaluate	Honest metrics + error analysis	Confusion matrix, precision, recall, F1	Beyond accuracy — identify failure modes

3. Technology Used

Python 3	Core programming language
pandas / NumPy	Data loading, cleaning, manipulation
scikit-learn	Preprocessing, Logistic Regression, Random Forest, metrics, threshold tuning
Matplotlib / Seaborn	Visualizations (distributions, churn rates, feature importance)
Streamlit	Interactive web app for live churn prediction (app.py)
Jupyter Notebook	Exploratory analysis and model development pipeline

4. Key Features

- 11 features used: CreditScore, Age, Tenure, Balance, NumOfProducts, EstimatedSalary (numeric), Geography & Gender (categorical), HasCrCard & IsActiveMember (binary). Target: Exited (20.4% churn).
- Geography is the strongest signal — Germany churn rate (32.4%) is ~2x France/Spain (~16%).
- Age > 50 and IsActiveMember = 0 strongly correlate with churn.
- Class imbalance handled via class_weight=balanced, not SMOTE (avoids synthetic data noise).
- Threshold tuned from default 0.5 to 0.554 using precision-recall curve on validation set — improved F1 without changing the model.
- Streamlit app (app.py) provides interactive predictions — deploys the trained model as a live tool.

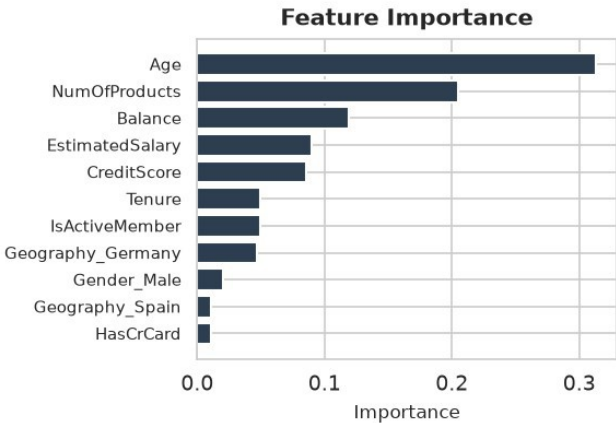
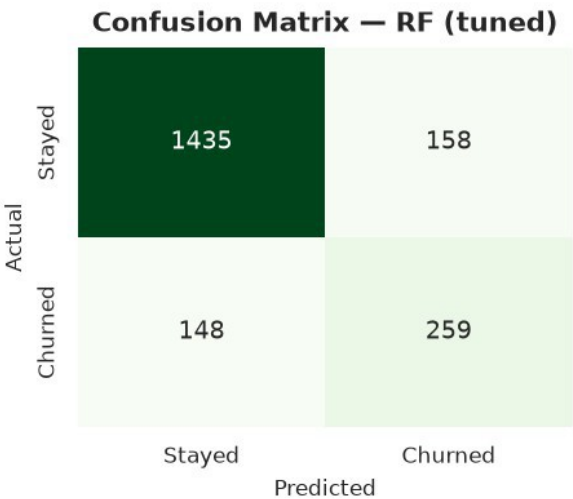
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Results, Error Analysis & Insights

5. Results & Model Comparison

Model	Precision	Recall	F1	ROC-AUC	Threshold
Logistic Regression	0.618	0.199	0.301	0.770	0.5
RF (default 0.5)	0.565	0.700	0.626	0.858	0.5

Best model: Random Forest with threshold = 0.554. Recall improved to 0.636 with precision 0.621, achieving F1 of 0.629. Logistic regression baseline achieved only 0.199 recall — too many churners missed.



6. Error Analysis — 3 Key Cases

Both models were tested on 2000 unseen customers. Logistic Regression missed most churners (recall 0.199). Random Forest captured significantly more (recall 0.636). Breakdown: TP=259, FP=158, FN=148. Below are the three most instructive cases.

Case A: False Positive (predicted churn, stayed)

Customer: Germany, Age 53, Balance 124K, Inactive, 1 product. Model predicted 95.3% churn — wrong.

The model over-indexes on the Germany + Inactive + High Balance combination. Business action: refine the model to separate true German churners from high-balance depositors who simply keep their accounts inactive.

Case B: False Negative (predicted stayed, churned)

Customer: France, Age 71, Balance 0, Active member, 2 products. Model predicted 94.1% stayed — wrong.

The model was fooled by active membership + low balance, missing the strong Age signal. The "silent churner" — elderly customers who leave without warning. Business action: flag elderly customers (>65) regardless of active status.

Case C: True Positive (predicted churn, churned)

Customer: Germany, Age 58, Balance 106K, Inactive, 4 products, CreditScore 546. Model: 98.1% churn — correct.

Textbook churn signature. The model correctly weighs the Geography + Age + Inactive combination. Business action: proactively contact similar profiles before they leave.

7. Key Takeaways

- Accuracy (85%) alone is misleading — null accuracy was 79.6%. Precision & recall tell the real story.
- Geography and Age are the strongest predictors. Germany and customers over 50 need separate models or feature engineering.
- Error analysis revealed blind spots: elderly low-balance customers (false negatives) and high-balance inactive German customers (false positives).
- The Streamlit app (app.py) makes the model usable by non-technical stakeholders — a deployable deliverable.